Lecture 20 (*April 7*)

Reading: Section 5.2
Viewing: Lecture 19

Formula (157) computes the determinant of a square matrix from its pivots, which is just what a computer would do. But there is a different way, namely the so-called "big formula" for the determinant, which is very important for understanding the theory and properties of the determinant.

Let's start from the following observation: determinants are linear functions of each row separately. As a formula, this means that for any numbers x_{ij} and $a_1, \dots, a_n$, $b_1, \dots, b_n$, we have:

$$\det \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_1 + b_1 & a_2 + b_2 & \dots & a_n + b_n \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nn} \end{bmatrix} = \det \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \dots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nn} \end{bmatrix} + \det \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_1 & b_2 & \dots & b_n \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nn} \end{bmatrix} \tag{163}$$

Very importantly, the matrix on the left is **NOT** the sum of the matrices on the right. Instead, all three matrices have the same elements on all rows except for the i–th one (that's the one where the a's and b's are in the formula), and i could be anything in $\{1, \dots, n\}$. Moreover, we have:

$$\det \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_1 & \lambda a_2 & \dots & \lambda a_n \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nn} \end{bmatrix} = \lambda \cdot \det \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \dots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nn} \end{bmatrix} \tag{164}$$

Remark. *Properties* (163) *and* (164) *also hold if we replace rows by columns (so the matrices in the left and right-hand sides would only differ in a single column) which makes sense given the fact that determinant is unchanged by transposing matrices.*

Here's an argument for (163). You can perform row exchanges in order to make $i = n$, i.e. the three matrices only differ on the last row. Perform Gaussian elimination on the three matrices in (163). It's not hard to see that the first $n - 1$ rows will look the same in all three matrices, so they will all have the same pivots $p_1, \dots, p_{n-1}$. When you finally get to do Gaussian elimination on the n–th row, denote the pivots on the n–th rows of the matrices in the right-hand side of (163) by α and β, respectively. Then a little thought shows that the pivot on the n–th row of the matrix in the left-hand side of (163) has to be $\alpha + \beta$. Then formula (157) implies that (163) boils down to:

$$\pm p_1 \dots p_{n-1}(\alpha + \beta) = \pm p_1 \dots p_{n-1}\alpha \pm p_1 \dots p_{n-1}\beta$$

which is definitely a true statement. Formula (164) is proved by a similar argument.

By applying formula (164) to all the rows of an $n \times n$ matrix, we get:

$$\boxed{\det(\lambda A) = \lambda^n \cdot \det A} \tag{165}$$

which you could also get by applying (156) to the matrix $B = \begin{bmatrix} \lambda & 0 & \dots & 0 \\ 0 & \lambda & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda \end{bmatrix}$.

Let's return to computing determinants. Since $\det(A+B) \neq \det A + \det B$ in general, we must resort to formula (163) in order to break up a general matrix into simpler, more manageable pieces. Let's take the 2×2 case for illustration. By applying (163) to the first row, we have:

$$\det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \det \begin{bmatrix} a_{11} & 0 \\ a_{21} & a_{22} \end{bmatrix} + \det \begin{bmatrix} 0 & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Then apply (163) to the second rows of each of the matrices in the right-hand side:

$$\det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \det \begin{bmatrix} a_{11} & 0 \\ a_{21} & 0 \end{bmatrix} + \det \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} + \det \begin{bmatrix} 0 & a_{12} \\ a_{21} & 0 \end{bmatrix} + \det \begin{bmatrix} 0 & a_{12} \\ 0 & a_{22} \end{bmatrix}$$

The first and fourth determinants in the right-hand side vanish because of (161), since they have full 0 columns. Therefore, we are left with:

$$\det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \det \underline{\begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix}} + \det \underline{\begin{bmatrix} 0 & a_{12} \\ a_{21} & 0 \end{bmatrix}} = a_{11}a_{22} - a_{12}a_{21}$$

Indeed, the first underlined matrix is already in row echelon form, so its determinant is the product of the two pivots. Meanwhile, the second underlined matrix would be in row echelon form as soon as we exchanged the two rows, which is responsible for the fact that its determinant has a -1 sign in front. Thus, we have proved formula (154).

In general, the procedure of successively applying (163) to the rows of the matrix leaves us with a sum of many determinants, all of which have a single non-zero entry on each row. Of all those determinants, the ones which have a full zero column will vanish, which means that the only surviving determinants are the ones where the non-zero elements are all on different rows, and also all on different columns. For example, in the 3×3 case, we have:

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \det \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix} + \det \begin{bmatrix} 0 & a_{12} & 0 \\ 0 & 0 & a_{23} \\ a_{31} & 0 & 0 \end{bmatrix} + \det \begin{bmatrix} 0 & 0 & a_{13} \\ a_{21} & 0 & 0 \\ 0 & a_{32} & 0 \end{bmatrix}$$

$$+ \det \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & 0 & a_{23} \\ 0 & a_{32} & 0 \end{bmatrix} + \det \begin{bmatrix} 0 & a_{12} & 0 \\ a_{21} & 0 & 0 \\ 0 & 0 & a_{33} \end{bmatrix} + \det \begin{bmatrix} 0 & 0 & a_{13} \\ 0 & a_{22} & 0 \\ a_{31} & 0 & 0 \end{bmatrix}$$

There are as many surviving determinants as possible permutations of the rows, and indeed, all of these determinants can be transformed into diagonal matrices by an appropriate row exchange. This leads to the following formula:

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31} \quad (166)$$

In general, the procedure gives a formula for the determinant as a sum over all permutations:

$$\{\sigma(1), \dots, \sigma(n)\} \quad \text{of} \quad \{1, \dots, n\}$$

and we have the following $\boxed{\textbf{big formula for the determinant}}$:

$$\det \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} = \sum_{\{\sigma(1),\dots,\sigma(n)\}}^{\text{permutations}} (-1)^{\operatorname{sgn}\sigma} a_{1\sigma(1)} a_{2\sigma(2)} \dots a_{n\sigma(n)} \tag{167}$$

The number sgn σ is called the **signature** of the permutation σ: it is the minimum number of successive (e.g. the i-th and $i+1$-th) row exchanges that you need to apply to the permutation matrix corresponding to σ, in order to turn it into a diagonal matrix.

Formula (167) is nice and explicit, but awful in practice. The number of summands for an $n \times n$ matrix is equal to the number of permutations of $\{1, \dots, n\}$, which is $n! = 1 \cdot 2 \cdot \dots \cdot n$. But there are matrices, particularly those which have a lot of zeroes, for which this formula is quite manageable. For example, if the i–th row only has a single non-zero entry (say on the j–th column) then we get:

$$\det \begin{bmatrix} \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{i-1,1} & \dots & a_{i-1,j-1} & a_{i-1,j} & a_{i-1,j+1} & \dots & a_{i-1,n} \\ 0 & \dots & 0 & a_{ij} & 0 & \dots & 0 \\ a_{i+1,1} & \dots & a_{i+1,j-1} & a_{i+1,j} & a_{i+1,j+1} & \dots & a_{i+1,n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \end{bmatrix} =$$

$$= \sum_{\substack{\{\sigma(1),\dots,\sigma(i-1),\sigma(i+1),\dots,\sigma(n)\} \\ \text{of } \{1,\dots,j-1,j+1,\dots,n\}}}^{\text{permutations}} (-1)^{\text{signature}} a_{1\sigma(1)} \dots a_{i-1,\sigma(i-1)} a_{ij} a_{i+1,\sigma(i+1)} \dots a_{n\sigma(n)}$$

$$= a_{ij} \cdot \underbrace{(-1)^{i+j} \det \begin{bmatrix} \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{i-1,1} & \dots & a_{i-1,j-1} & a_{i-1,j+1} & \dots & a_{i-1,n} \\ a_{i+1,1} & \dots & a_{i+1,j-1} & a_{i+1,j+1} & \dots & a_{i+1,n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \end{bmatrix}}_{\text{cofactor } C_{ij}} \tag{168}$$

This is because the only terms which give a non-zero contribution to (167) are those with $\sigma(i) = j$. So computing an $n \times n$ determinant whose only non-zero entry on row i is on column j boils down to computing the determinant of the $(n-1) \times (n-1)$ matrix obtained by removing row i and column j.

Formula (168) leads to a general way of computing determinants, known as **cofactor expansion**. Explicitly, for a matrix $A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix}$ and any row i, we have:

$$\det A \overset{(163)}{=} \sum_{j=1}^{n} \det \begin{bmatrix} \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{i-1,1} & \dots & a_{i-1,j-1} & a_{i-1,j} & a_{i-1,j+1} & \dots & a_{i-1,n} \\ 0 & \dots & 0 & a_{ij} & 0 & \dots & 0 \\ a_{i+1,1} & \dots & a_{i+1,j-1} & a_{i+1,j} & a_{i+1,j+1} & \dots & a_{i+1,n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \end{bmatrix}$$

We may invoke formula (168) to compute the right-hand side, and we obtain:

$$\boxed{\det A = a_{i1} \cdot C_{i1} + \cdots + a_{in} \cdot C_{in}} \tag{169}$$

where C_{ij} is called the (i, j) **cofactor**:

$$\boxed{C_{ij} = (-1)^{i+j} \det M_{ij}} \tag{170}$$

where M_{ij} is the matrix obtained by removing row i and column j from A.

Remark. *The cofactor expansion can also be done along columns instead of rows:*

$$\boxed{\det A = a_{1j} \cdot C_{1j} + \cdots + a_{nj} \cdot C_{nj}} \tag{171}$$

for all $j \in \{1, \ldots, n\}$.

Let's do an example, by computing the determinant of the matrix:

$$A = \begin{bmatrix} 7 & 0 & 3 & -1 \\ 0 & 4 & 3 & 0 \\ 2 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 \end{bmatrix}$$

This matrix has a lot of zeroes, so it's perfect for cofactor expansion. To keep your work as light as possible, pick a row with a lot of zeroes. For example, cofactor expansion for the third row gives:

$$\det A = 2 \cdot (-1)^{3+1} \det \begin{bmatrix} 0 & 3 & -1 \\ 4 & 3 & 0 \\ 2 & 0 & 1 \end{bmatrix} + 1 \cdot (-1)^{3+4} \det \begin{bmatrix} 7 & 0 & 3 \\ 0 & 4 & 3 \\ 0 & 2 & 0 \end{bmatrix}$$

We can compute the two determinants in the right-hand side by cofactor expansion for the first and third rows, respectively:

$$\det A = 2 \cdot (-1)^{3+1} \left(3 \cdot (-1)^{1+2} \det \begin{bmatrix} 4 & 0 \\ 2 & 1 \end{bmatrix} + (-1) \cdot (-1)^{1+3} \det \begin{bmatrix} 4 & 3 \\ 2 & 0 \end{bmatrix} \right) +$$

$$+1 \cdot (-1)^{3+4} \cdot 2 \cdot (-1)^{3+2} \det \begin{bmatrix} 7 & 3 \\ 0 & 3 \end{bmatrix} = 2((-3) \cdot 4 + (-1) \cdot (-6)) + 2 \cdot 21 = 30$$

Just to do a sanity check, let's also compute the determinant of A by cofactor expansion along a column (say the second column) and hope that we get the same answer:

$$\det A = 4 \cdot (-1)^{2+2} \det \begin{bmatrix} 7 & 3 & -1 \\ 2 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} + 2 \cdot (-1)^{2+4} \det \begin{bmatrix} 7 & 3 & -1 \\ 0 & 3 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

Now let's compute the two determinants in the right-hand side by cofactor expansion for the second and third column, respectively:

$$\det A = 4 \cdot (-1)^{2+2} \left(3 \cdot (-1)^{1+2} \det \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \right) + 2 \cdot (-1)^{2+4} \left((-1) \cdot (-1)^{1+3} \det \begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix} + \right.$$

$$\left. + 1 \cdot (-1)^{3+3} \det \begin{bmatrix} 7 & 3 \\ 0 & 3 \end{bmatrix} \right) = 4 \cdot (-3) \cdot 2 + 2((-1) \cdot (-6) + 1 \cdot 21) = 30$$